

Welcome to Machine Learning

What is Machine Learning?

- **Definition:** Machine learning is the science of getting computers to **learn without being explicitly programmed**.
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Real-World Applications of Machine Learning

Machine learning is a pervasive technology used in numerous applications, often without the user's explicit awareness.

Consumer Applications

- **Web Search:** Ranking search results (e.g., Google, Bing).
- **Social Media:** Automatically recognizing and tagging friends in photos (e.g., Instagram, Snapchat).
- **Recommendation Engines:** Suggesting movies, products, or music based on user preferences (e.g., video streaming services).
- **Voice Assistants:** Powering voice-to-text and digital assistants (e.g., Siri, Google Assistant).
- **Email:** Filtering spam messages from the inbox.

Industrial & Societal Applications

- **Energy:** Optimizing power generation from wind turbines.
- **Healthcare:** Assisting doctors in making more accurate medical diagnoses.
- **Manufacturing:** Using computer vision to inspect products for defects on an assembly line.